

# Embedded Systems

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## Computing and Mathematics

May, 2026



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|---------------------|--------------------------------|----------------------------|
| <b>Short Title:</b> | Embedded Systems               |                            |
| <b>Faculty</b>      | Science and Computing          |                            |
| <b>Department</b>   | Computing and Mathematics      |                            |
| <b>Cluster</b>      | Automotive, Automation and IoT |                            |
| <b>Author</b>       | BJACKMAN                       |                            |
| <b>Credits</b>      | 5                              | <b>Level:</b> Introductory |

## Description of Module / Aims

In this module students will learn how to develop embedded real-time software applications for modern microcontrollers intended for the control of industrial processes and mechatronic systems.

## Programmes

|           |                                              |           |
|-----------|----------------------------------------------|-----------|
| COMP-0195 | BSc (Hons) in Applied Computing (WD_KACCM_B) | 2 / 3 / E |
| COMP-0195 | BSc (Hons) in Applied Computing (WD_KCOMP_B) | 2 / 3 / E |
| COMP-0195 | BSc (Hons) in Computer Science (WD_KCMSC_B)  | 2 / 3 / E |

## Indicative Content

- C Programming basics: Language features; Functions; Data Types; Structures; Pointers; Bit Manipulation
- Interfacing Principles: I/O types; Memory-mapped I/O; Control and Status Registers
- Digital I/O: Direction and Data Registers; Switch debouncing; Output control
- Timers and Counters
- Analogue Inputs: Sampling theorem; Analogue-to-Digital Converter
- Analogue Outputs: Pulse Width Modulation (PWM)
- Microcontroller architecture

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Define the interfacing requirements for controlling an industrial process.
2. Apply a control system software design to a common micro-controller.
3. Code a microcontroller software system using C.
4. Test and debug embedded control software on common microcontrollers.

## Learning and Teaching Methods

- Combination of lectures and lab-based practicals.
- The lectures will cover the theory and underlying technologies in embedded application development.
- The lab-based practicals, building on the theoretical knowledge from the lectures, provide the practical skills to develop and test microcontroller software using a modern microcontroller.
- Students will be encouraged to enhance their lab work and assessment submissions using self-directed research and learning into broader embedded systems topics.

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 50%      | 1,2               |
| Continuous Assessment     | 50%      | 2,3,4             |
| <i>Assignment</i>         | 25%      | 2,3,4             |
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## Assessment Criteria

**<40%:** Inability to develop simple software designs from user requirements.

**40%-49%:** Able to specify interfacing and software requirements for simple discrete control systems.

**50%-59%:** All the above and in addition be able to implement microcontroller-based solutions.

**60%-69%:** In addition, be able to develop optimised designs for a common micro-controller.

**70%-100%:** All previous to an excellent level. Be able to demonstrate an awareness of software quality and performance considerations when developing embedded software applications.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 12        |           |
| Practical            | 36        |           |
| Independent Learning | 87        |           |

## Supplementary Material

- Qian, K., D. Den Haring and L. Cao. \_Embedded Software Development with C\_. New York: Springer, 2009.

## Requested Resources

- COMPUTER LAB: BYOD Lab